

YouTube Comment Toxicity & Hate Speech Detector

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Introduction

YouTube has become an international platform for users to share content and build online communities. The comment section functions as an essential component of this ecosystem because it enables creators and viewers to interact directly with one another. The platform's expanding user base makes it more challenging to maintain a secure and safe platform environment. The platform requires automatic Natural Language Processing (NLP) systems because their current system cannot handle the daily influx of millions of comments which demand manual review.



Problem Statement

The expansion of digital social networks has brought about an increase in harmful conduct which includes both harassment and hate speech. The harmful comments which people make can be divided into different types which include obscenity and threats and identity-based hate. These different types of harmful comments make it impossible for the community to maintain its well-being.

The primary technical challenges include:

- **Multi-label Classification:** The process demands accurate identification of one comment which belongs to multiple harmful categories. The categories to be identified include toxic and severe toxic and obscene and threat and insult and identity hate.
- **Data Imbalance:** The process involves managing extensive datasets which contain "normal" comments that greatly exceed the quantity of "abnormal" comments. The training dataset used for this study includes 143346 normal comments out of a total 159571 comments.
- **Scalability:** The process needs to create a model which can effectively process text data that contains 10000 dimensions of information. The 10,000 features extracted through TF-IDF vectorization represent the high-dimensional text data which needs to be processed by the model.

Project Objectives

The researchers created an automated moderator system which uses natural language processing to automatically categorize YouTube comments, thus decreasing the need for human content moderators. The system needs to detect six different types of toxicity which include toxic, severe toxic, obscene, threat, insult, and identity hate.

The system needs to detect rare critical categories which include "threat" and "identity hate" through the implementation of class weighting and other strategies. The researchers aim to enhance model performance by developing a system which processes extensive data more swiftly while maintaining high accuracy as measured by AUC-ROC.

Methodology

- **Data Collection:** Utilize the Jigsaw Toxic Comment dataset containing over 159,000 labeled samples.
- **Text Preprocessing:**
 - Clean noise (URLs, special characters, and numbers).
 - Apply SpaCy lemmatization and remove stopwords to focus on core word meanings.
- **Feature Engineering:** Convert processed text into numerical vectors using TF-IDF with a maximum of 10,000 features.
- **Model Training:**
 - Implement a Multi-Output Classifier to handle multiple labels per comment.
 - Utilize Weighted Linear SVM as the primary algorithm to address the 143k+ majority of normal comments versus toxic ones.
- **Evaluation:** Measure success using AUC-ROC and Weighted F1-Score to ensure reliable performance across all categories.

Results

- **High Discriminatory Power:** The system is expected to achieve an AUC-ROC score of 0.9948, demonstrating an excellent ability to distinguish between normal and toxic comments.
- **Effective Imbalance Handling:** By implementing a weighted approach, the model will reach a Weighted F1-Score of 0.7482,

approach, the model will reach a Weighted F1-Score of 0.7482, ensuring reliable detection even for minority classes like "threat" and "identity hate".

- **Real-time Performance:** The optimized Linear SVM architecture will provide fast inference times, making it suitable for processing large volumes of YouTube comment data.
- **Ready-to-use Model:** The project will deliver a finalized and saved model (`toxic_weighted_linear_svm_model.pkl`) along with its TF-IDF vectorizer for immediate deployment.

Future Work & Development

- **YouTube Integration** The system will achieve integration with YouTube Data API v3 which will enable automatic retrieval of live video comments from designated video URLs for use in real-time analysis.
- **Advanced Deep Learning Models** The system will use Transformer-based architectures which include BERT and RoBERTa as its new detection system because the current Weighted Linear SVM baseline fails to identify complex contextual information and ironic and sarcastic expressions.
- **Multilingual Support** The model will extend its language capabilities to include multiple languages while focusing on Arabic dialects to better assist the local and regional YouTube creator community.
- **Model Optimization** The team will optimize TF-IDF features and hyper-parameters until the model achieves high AUC-ROC score (0.9948) but decreases its operational size which enables quicker analysis during busy periods.